Homework 13: 10/22 Content

MATH 1271, Garrett Credi

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Problems listed as AC are found in the Active Calculus textbook at the end of the listed topic. You should be finding these in the *Exercises* section of the page.

Problems listed as APEX are found in the APEX Calculus Textbook again at the end of the listed topic. You should be finding these in the *Exercises* section of the page.

Problems listed as PAULS-PRACT are found in Paul's Online Math Notes - Review Problems while problems listed as PAULS-ASSIGN are found in Paul's Online Math Notes - Assignment Problems and are indexed via [Topic]-[Section]-[Problem Number]

1 Type 1 - Related Rates

Choose 8 of the following problems to do.

1. AC 3.1.1-12 (each of these counts as one problem)
2. APEX 4.2.3-15 (each of these counts as one problem)
3. PAULS-ASSIGN.Derivatives.Related Rates.1-25 (each of these counts as one problem)

2 Challenge Problem

Suppose $f(x)$ is some differentiable function and set $g(x) = \ln(f(x))$. What is the derivative of $g(x)$ in terms of $f(x)$ and $f'(x)$? Use what you find to compute the derivative of the function $h(x) = x^x$.